

Deepfake Image Detection Using Frequency-Domain Analysis and Explainable AI

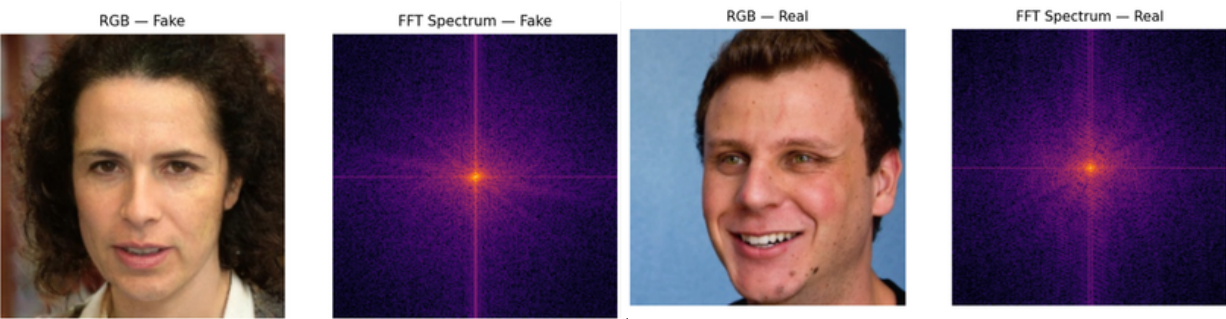
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01 INTRODUCTION

- Deepfakes are realistic AI-generated images
- Used in fraud and misinformation
- Hard to detect visually

02 MOTIVATION

- **RGB** misses hidden artifacts
- **FFT** reveals frequency inconsistencies
- Combining both improves detection



03 CONTRIBUTIONS

- Compared **RGB**, **FFT**, **Hybrid** models
- Analyzed frequency-domain features
- Used Grad-CAM for explainability
- Studied failure cases

04 DATASET

- 30,000 images (15k real / 15k fake)
- 70% train / 15% val / 15% test split
- Image size: 224×224

Sample Real vs Fake Faces



05 METHODOLOGY

Exp 1 — RGB Baseline

ResNet18 (ImageNet pretrained), fine-tuned for binary Real/Fake classification.

Exp 2 — FFT Model

ResNet18 (random init). Input: per-channel log-magnitude FFT spectrum.

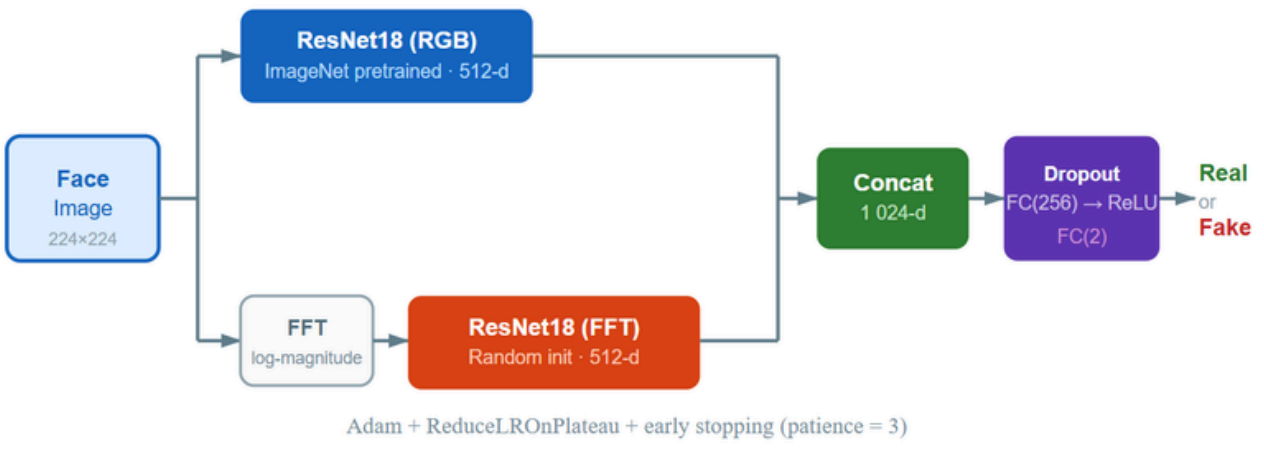
Exp 3 — Hybrid

Dual-branch: RGB (pretrained) + FFT (random) → concat 1024-d → FC classifier.

SETTING	RGB	FFT	HYBRID
Pretrained	ImageNet	None	RGB: yes / FFT: no
Feature dim	512-d	512-d	1 024-d
Learning rate	1e-4	1e-3	5e-5

06 HYBRID MODEL ARCHITECTURE

- Two parallel branches: **RGB** and **FFT**
- Each uses ResNet18 (512-d features)
- Features concatenated (1024-d)
- Passed through FC layers for classification



08 CONCLUSION

- **FFT** alone is insufficient — 61% accuracy barely exceeds chance; frequency cues lack spatial context
- **RGB** baseline is strong — ImageNet features transfer well, AUC = 0.9975
- **Hybrid** wins across all metrics — perfect recall (100%) and AUC = 0.9998; spatial + spectral cues are complementary

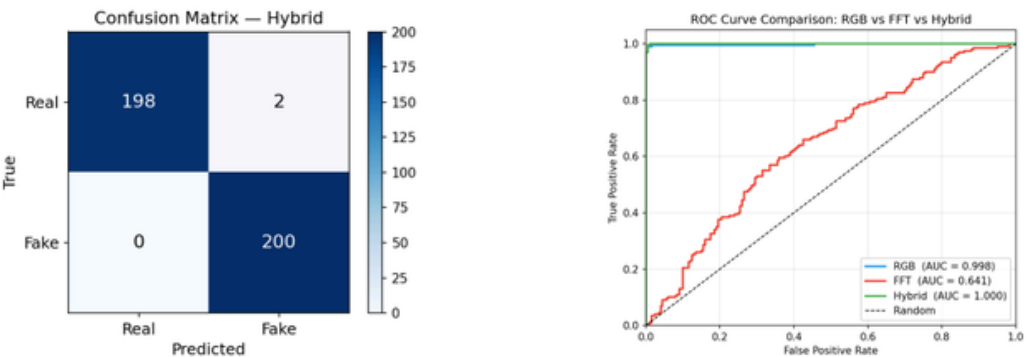
07 RESULTS

Quantitative Results

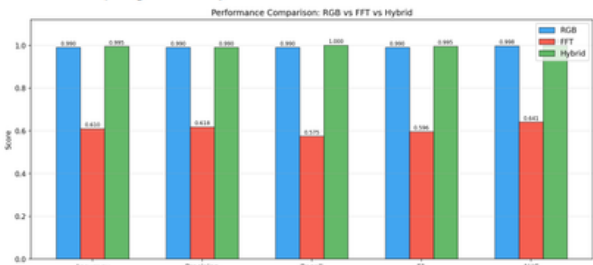
	99.0% RGB ACC.	61.0% FFT ACC.	99.5% HYBRID ACC.	
MODEL	ACCURACY	PRECISION	RECALL	AUC
RGB	99.00%	99.00%	99.00%	0.9975
FFT	61.00%	61.83%	57.50%	0.6412
Hybrid ★	99.50%	99.01%	100.0%	0.9998

- ⚠ **FFT alone insufficient** — 61% accuracy barely exceeds chance; frequency cues lack spatial context.
- ▶ **RGB baseline is strong** — ImageNet features transfer well, AUC = 0.9975.
- ✓ **Hybrid wins across all metrics** — perfect recall (100%) and AUC = 0.9998; spatial + spectral cues are complementary.

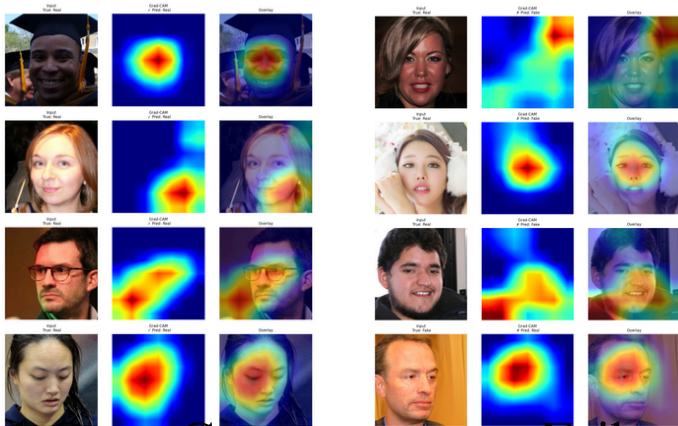
Visual Results



Confusion Matrix (Hybrid) ROC Curves — RGB vs FFT vs Hybrid



Accuracy · Precision · Recall · F1 · AUC — all models



Grad-Cam: Correct cases vs Failure Cases